



III. Solutions and Answers

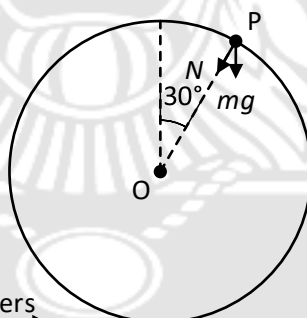
The suggested solutions and answers below are only for your reference and have not been verified to be error-free. If you have any doubt, please discuss with your tutor to clarify.

MCQs

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
C	C	B	B	C	C	B	B	C	C	D	D	A	C	B

16	17	18	19	20	21	22	23	24	25					
B	A	A	D	A	A	D	D	D	C					

1 C



Resolving the forces acting on the roller coaster at P radially,

$$mg \cos 30^\circ + N = \frac{mv^2}{r}$$

When it loses contact, $N = 0$.

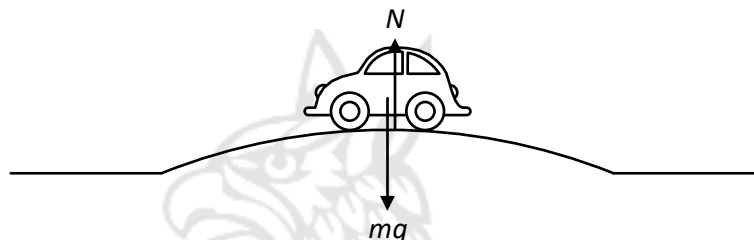
$$v = \sqrt{rg \cos 30^\circ} = \sqrt{1.0 \times 9.81 \times \cos 30^\circ} = 2.9 \text{ m s}^{-1}$$

2 C Resultant force towards the centre of the hump provides the centripetal force (see MCQ Q3).

$$mg - N = \frac{mv^2}{r}$$

$$\Rightarrow v = \sqrt{\frac{r}{m}(mg - N)} \leq \sqrt{rg} = \sqrt{3.0 \times 9.81} = 5.4 \text{ m s}^{-1}$$

So maximum speed of electric scooter is 5.4 m s^{-1} .

3 B

Resultant force towards the centre of the hump provides the centripetal force.

$$mg - N = \frac{mv^2}{r}$$

Car loses contact with road when $N = 0$.

$$v = \sqrt{rg} = \sqrt{30 \times 9.81} = 17.2 \text{ m s}^{-1}$$

4 B Apparent weight is normal contact force which provides for the centripetal force.

$$N = F_c$$

$$\Rightarrow mg = mr\omega^2$$

Since $g \propto r$ as ω^2 is constant.

$$R_2 = \frac{g_2}{g_1} \times R_1 = \frac{3.72}{9.81} \times 2150 = 800 \text{ m}$$

Note: g is the simulated acceleration due to gravity.

5 C Linear velocity

$$v = r\omega = 6.38 \times 10^6 \cos 31.2^\circ \times \frac{2\pi}{24 \times 60 \times 60} = 397 \text{ m s}^{-1}$$

6 C Since speed $v = r\omega$,

$$\frac{\text{indicated speed} - \text{actual speed}}{\text{actual speed}} \times 100\% = \frac{R\omega - r\omega}{r\omega} \times 100\% = \frac{61 - 51}{51} \times 100\% = +20\%$$

7 B Resolving forces vertically,

$$N \sin \theta = mg \quad (1)$$

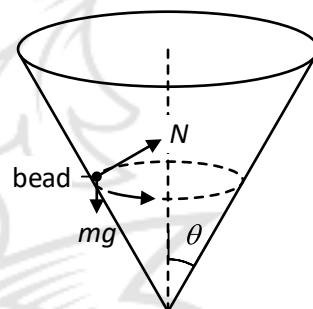
Horizontally,

$$N \cos \theta = ma_c = mg \quad (2)$$

(1)/(2):

$$\tan \theta = 1$$

$$\Rightarrow \theta = 45^\circ$$



- 8 B** As ω increases, the frictional force on each disc will increase (up to its respective $f_{\max}$) to provide for the increasing centripetal force ($mr\omega^2$) required. When $f_{\max} = mr\omega^2$, the disc is on the verge of slipping off.

The angular velocity ω of the turntable when each disc begins to slip are

$$\omega_1 = \sqrt{\frac{f_{\max}}{MR}}$$

$$\omega_2 = \sqrt{\frac{f'_{\max}}{4MR}} = \sqrt{\frac{2f_{\max}}{4MR}} = \sqrt{\frac{f_{\max}}{2MR}} < \omega_1$$

Hence, disc 2 will experience maximum frictional force and slip off first.

- 9 C** To complete the vertical circle, the pendulum and bullet must be able to reach its highest point with a minimum speed of zero (minimum speed of $\sqrt{gL}$ if it is suspended from a string).

By the principle of conservation of energy,

loss in K.E. = gain in G.P.E.

$$\Rightarrow \frac{1}{2}(m+M)v^2 - 0 = (m+M) \times g \times 2L$$

$$\Rightarrow v = 2\sqrt{gL}$$

By the principle of conservation of linear momentum,

$$mu = (m+M)v$$

$$u = \frac{(m+M)v}{m} = \frac{2(m+M)\sqrt{gL}}{m}$$

- 10 C** Since the drive wheel and the turntable are in contact, they have the same linear speeds, i.e., $V = v$. Angular velocities are related:

$$v = R\Omega = r\omega \Rightarrow \Omega = \left(\frac{r}{R}\right)\omega$$

- 11 D** Centripetal acceleration

$$a = r\omega^2 = r\left(\frac{2\pi}{T}\right)^2 = \frac{4\pi^2 r}{T^2} \Rightarrow T = 2\pi\sqrt{\frac{r}{a}} \Rightarrow 10T = 20\pi\sqrt{\frac{r}{a}}$$

- 12 D** Resolving forces horizontally,

$$F \sin \theta = m(3.0 + 5.0 \sin 30^\circ)\omega^2 \quad (1)$$

Vertically,

$$F \cos \theta = mg \quad (2)$$

(1)/(2):

$$\tan 30^\circ = \frac{(3 + 5 \sin 30^\circ)4\pi^2}{9.81T^2} \Rightarrow T = 6.2 \text{ s}$$

- 13 A At the top,

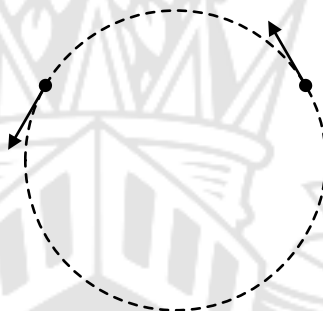
$$mg + N = \frac{mv^2}{r}$$

The normal contact force $N \geq 0$ for ball to remain inside pail.

$$\frac{mv^2}{r} \geq mg \Rightarrow v \geq \sqrt{rg} = \sqrt{0.70 \times 9.81} = 2.6 \text{ m s}^{-1}$$

- 14 C The horizontal component of tension provides the leftward centripetal force for the pendulum to move in a circular path.

- 15 B



Although the speed of stone is the same at the same height, its velocity depends on direction of motion and hence may be different.

- 16 B Let L be the lift acting on the bird. Resolving forces horizontally,

$$L \sin 24^\circ = \frac{mv^2}{r} \quad (1)$$

Vertically,

$$L \cos 24^\circ = mg \quad (2)$$

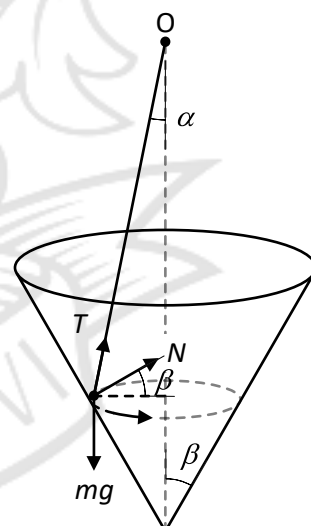
(1)/(2):

$$\tan 24^\circ = \frac{v^2}{rg}$$

$$v = \sqrt{2.0 \times 9.81 \times \tan 24^\circ} = 3.0 \text{ m s}^{-1}$$

- 17 A The free-body diagram shows the forces acting on the bob. The centripetal force on the bob is provided by the horizontal components of N and T .

$$F_c = T \sin \alpha + N \cos \beta$$



- 18 A Frictional force on each disc provides for the centripetal force.

$$f = mr\omega^2$$

So

$$\frac{f_{3.0}}{f_{6.0}} = \frac{3.0 \times 2.0 \times \omega^2}{6.0 \times 5.0 \times \omega^2} = 0.20$$

- 19 D At its highest point of the vertical circular motion, the drone experiences centripetal acceleration (downward) and a tangential acceleration (rightward).

The acceleration of the drone is the vector sum of centripetal and tangential accelerations.

- 20 A Since X and Y are on the same turntable, the angular speeds at X and Y are equal.

Since linear speed $v = r\omega$ and $r_X > r_Y$, $v_X > v_Y$.

Since $r_X > r_Y$, the centripetal force $F_c = mr\omega^2$ required to keep the sand in circular motion is greater at X. Since frictional force f provides for the centripetal force, the sand at X will fly off first as the frictional force at X will exceed the maximum frictional force first.

- 21 A Angular displacement of car in 4.0 s is

$$\theta = 4.0 \times 0.5\pi = 2\pi \text{ rad.}$$

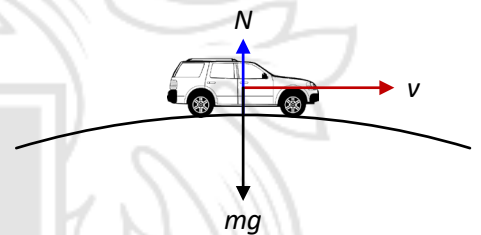
So the car is back at P after 4.0 s. Therefore, its displacement from P is zero.

- 22 D Resultant force provides for centripetal force.

$$mg - N = \frac{mv^2}{r}$$

Force exerted on car by bridge

$$N = mg - \frac{mv^2}{r}$$



- 23 D Let m be the mass of the aircraft.

The forces acting on the plane are lift L and weight mg .

The centripetal force is provided by the horizontal component of lift.

$$L \sin 32^\circ = \frac{mv^2}{r} \quad \text{①}$$

The weight is balanced by the vertical component of lift.

$$L \cos 32^\circ = mg \quad \text{②}$$

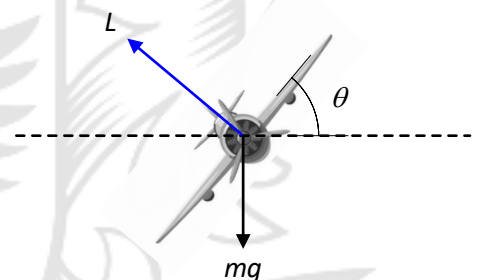
① ÷ ②:

$$\tan 32^\circ = \frac{v^2}{rg}$$

For radius $r/2$ and speed $2v$,

$$\tan \theta = \frac{(2v)^2}{(r/2)g} = \frac{4}{1/2} \left(\frac{v^2}{rg} \right) = 8 \times \tan 32^\circ$$

$$\Rightarrow \theta = 79^\circ$$

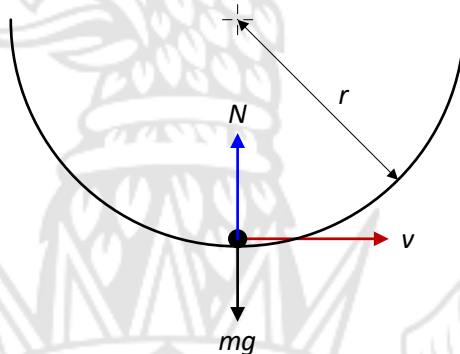


- 24 D By principle of conservation of energy,

gain in K.E. = loss in G.P.E.

$$\frac{1}{2}mv^2 - 0 = mgr$$

$$\Rightarrow \frac{mv^2}{r} = 2mg$$



At the bottom of the bowl, centripetal force is provided by the resultant force.

$$N - mg = \frac{mv^2}{r}$$

$$\Rightarrow N = \frac{mv^2}{r} + mg = 2mg + mg = 3mg$$

Note: The size of the object is considered to be small so the radius of circular path is also r .

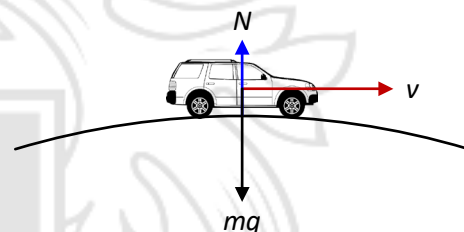
- 25 C At the highest point of the hump, the centripetal force is provided by the resultant force.

$$mg - N = \frac{mv^2}{r}$$

The driver experiences weightlessness when normal contact force N on driver is zero.

$$mg - 0 = \frac{mv^2}{r}$$

$$v = \sqrt{rg}$$



Note: Although the forces shown are acting on the car, the analysis is the same for the driver.

Structured Questions

1. [AJC/H2/Prelims 2018/P2/2]

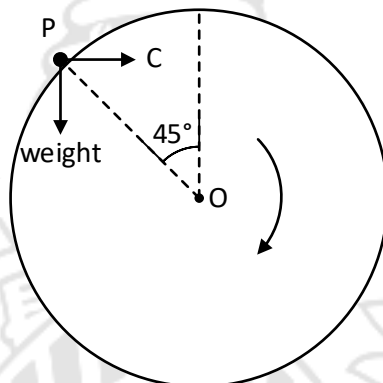
- a. Angular velocity of disc

$$\omega = 2\pi \left(\frac{N}{t} \right) = 2\pi \left(\frac{100}{60} \right) = 10.5 \text{ rad s}^{-1}$$

Centripetal force

$$F = mr\omega^2 = 3.8 \times 10^{-3} \times 0.118 \times 10.5^2 = 0.049 \text{ N}$$

- b.



Arrow C must be equal in length and perpendicular to weight.

- c. i. At the lowest point, the contact force (upwards) and the weight (downwards) act opposite in direction. Since the resultant force provides for the centripetal force (upwards towards O), the contact force will be greatest at the lowest point. Therefore the plasticine is most likely to first lose contact with the disc at the lowest point.

- ii. At the lowest point,

$$C_{\max} - mg = mr\omega^2$$

$$\omega = \sqrt{\frac{C_{\max} - mg}{mr}} = \sqrt{\frac{0.23 - (3.8 \times 10^{-3} \times 9.81)}{3.8 \times 10^{-3} \times 0.118}} = 21 \text{ rad s}^{-1}$$

2. [CJC/H2/Prelims 2018/P3/1]

- a. i. As the wrench falls, it loses gravitational potential energy and gains kinetic energy. Therefore, its speed increases.

Since the centripetal acceleration is directly proportional to the square of the velocity for constant radius of the circular path, the centripetal acceleration of the wrench increases as it slides from A to B.

- ii. $N = 0$

- b. Since $N = 0$ when the wrench loses contact with the surface of the dome, resolving the forces radially,

$$mg \cos \theta = \frac{mv^2}{r}$$

$$mg \left(\frac{h}{R} \right) = \frac{mv^2}{R} \Rightarrow h = \frac{v^2}{g}$$

- c. i. 1. From b.,

$$v = \sqrt{hg} = \sqrt{\frac{2}{3} \times 30.0 \times 9.81} = 14.0 \text{ m s}^{-1}$$

2. From the diagram,

$$\cos \theta = \frac{h}{R} \Rightarrow \theta = \cos^{-1} \left(\frac{h}{R} \right) = \cos^{-1} \left(\frac{2}{3} \right) = 48.2^\circ$$

- ii. Considering vertical motion and taking vectors downwards as positive,

$$s_y = u_y t + \frac{1}{2} g t^2$$

$$\Rightarrow \frac{2}{3} \times 30.0 = v \sin \theta \times t + \frac{1}{2} \times 9.81 \times t^2 = (14.0 \sin 48.2^\circ) t + \frac{1}{2} \times 9.81 \times t^2$$

Solving,

$$t = 1.22 \text{ s}$$

- iii. Horizontal distance

$$x = u_x t = v \cos \theta \times t = 14.0 \cos 48.2^\circ \times 1.22 = 11.4 \text{ m}$$

3. [DHS/H2/Prelims 2018/P3/4]

- a. i. The radian is the angle θ subtended by an arc of length s equal to the radius r of the arc, i.e.,

$$\theta = \frac{s}{r}$$

- ii. The arc length of one complete revolution is the circumference of a circle, i.e., $s = 2\pi r$.

$$\theta = \frac{s}{r} = \frac{2\pi r}{r} = 2\pi$$

- b. i. The lowest point.

- ii. The resultant force provides the centripetal force,

$$T - mg = mr\omega^2$$

$$\Rightarrow \omega = \sqrt{\frac{T - mg}{mr}} = \sqrt{\frac{18.0 - 3.0}{(3.0/9.81) \times 0.85}} = 7.6 \text{ rad s}^{-1}$$

- c. i. Since the mass is in equilibrium vertically,

$$T \cos \theta = mg \Rightarrow T = \frac{mg}{\cos \theta} = \frac{3.0}{\cos 35^\circ} = 3.7 \text{ N}$$

- ii. The resultant force acting on the stone is horizontal, towards the centre of its circular path. This is provided by the horizontal component of the tension.

$$F_R = T \sin \theta = 3.7 \sin 35^\circ = 2.1 \text{ N}$$

4. [JJC/H2/Prelims 2018/P3/3]

- a. i. When mass is in equilibrium at the lowest position,

$$mg = kx \Rightarrow k = \frac{mg}{x} = \frac{0.50 \times 9.81}{0.148 - 0.130} = 273 \text{ N m}^{-1}$$

- ii. Above C, only weight provides the centripetal force.

$$mg = mr\omega^2 \Rightarrow \omega = \sqrt{\frac{g}{r}} = \sqrt{\frac{9.81}{0.130}} = 8.69 \text{ rad s}^{-1}$$

- iii. At the lowest point, resultant force provides the centripetal force.

$$T - mg = mr\omega^2$$

$$\Rightarrow 272.5x - 0.50 \times 9.81 = 0.50 \times (0.13 + x) \times 8.7^2$$

$$\Rightarrow x = 0.0418 \text{ m}$$

$$\Rightarrow L = 0.130 + 0.0418 = 0.172 \text{ m}$$

- b. i. 1. The centripetal force experienced by the mass at the equator,

$$F_c = m r \omega^2 = 5.00 \times 6.40 \times 10^6 \times \left(\frac{2\pi}{24 \times 60 \times 60} \right)^2 = 0.169 \text{ N}$$

2. The resultant force provides the centripetal force.

$$W - N = F_c \Rightarrow N = 5.00 \times 9.81 - 0.169 = 48.9 \text{ N}$$

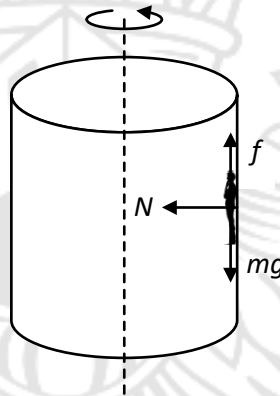
- ii. At the poles, the mass is not accelerating. Hence normal contact force is equal in magnitude but opposite in direction to gravitational force.

At the equator, the resultant force ($W - N$) cannot be zero since the mass is undergoing circular motion about centre of Earth.

Note: It can be said that a fraction of the gravitational force provides the centripetal force.

5. [NYJC/H2/Prelims 2018/P3/3]

- a. i.



f is friction; mg is weight, N is normal contact force.

- ii. Since the passenger is in equilibrium vertically,

$$f = mg \quad \text{①}$$

Normal contact force provides the centripetal force.

$$N = \frac{f}{\mu} = \frac{mv^2}{R} \quad \text{②}$$

② ÷ ①:

$$\frac{1}{\mu} = \frac{v^2}{Rg} \Rightarrow v = \sqrt{\frac{Rg}{\mu}}$$

Hence velocity v is independent of mass m .

- b. i. At point A, resultant force provides for centripetal force.

$$mg - N = \frac{mv^2}{r}$$

To remain in contact with the track, $N \geq 0$.

$$\frac{mv^2}{r} \leq mg \Rightarrow v \leq \sqrt{rg} = \sqrt{95 \times 9.81} = 30.5 \text{ m s}^{-1}$$

- ii. By principle of conservation of energy,

loss in G.P.E. = gain in K.E.

$$mgh + \frac{1}{2}mv^2 = mg \times 2 \times 95 + \frac{1}{2}mv^2$$

$$h = 206 \text{ m}$$

6. [PJC/H2/Prelims 2018/P2/2]

- a. A body moving in a circular path with constant speed has constantly changing direction of motion. Its velocity is constantly changing, and it must experience an acceleration towards the centre of the circular path.

b. i.



N is normal contact force, mg is weight.

ii. At Y,

$$mg + N = \frac{mv^2}{R}$$

For the ball to be just in contact with the loop, $N = 0$.

$$mg = \frac{mv^2}{R} \Rightarrow v = \sqrt{gR}$$

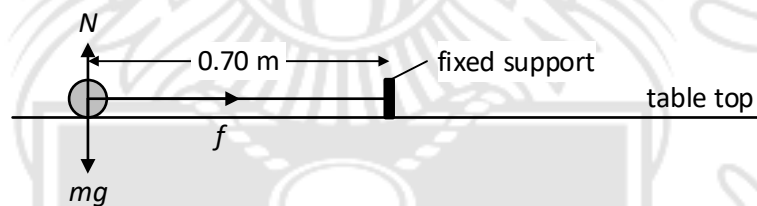
iii. By principle of conservation of energy,

$$mgH + \frac{1}{2}mv^2 = mg \times 2R + \frac{1}{2}mv^2$$

$$mgH + 0 = mg \times 2R + \frac{1}{2}mgR \Rightarrow H = 2.5R$$

7. [RVHS/H2/Prelims 2018/P3/3]

a.

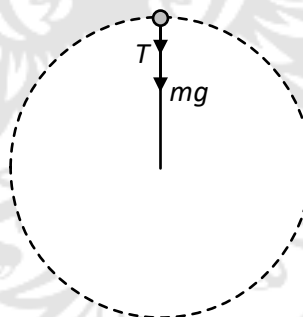


T is tension, mg is weight, N is normal contact force.

b. The tension in the string provides the centripetal force.

$$kx = \frac{mv^2}{r} \Rightarrow v = \sqrt{\frac{kx}{m}} = \sqrt{\frac{40 \times 0.20 \times 0.70}{0.050}} = 11 \text{ m s}^{-1}$$

c. Consider the mass moving in a circular path in a vertical plane.



$$mg + T = \frac{mv^2}{r}$$

where T is tension in the string and mg is the weight of the mass. Since tension in string $T \geq 0$,

$$mg \geq \frac{mv^2}{r} \Rightarrow v \leq \sqrt{rg} = 3.13\sqrt{r}$$

So minimum speed $v_{\min} = 3.13\sqrt{r}$.

8. [VJC/H2/Prelims 2018/P3/8(part)]

- a. i. By the principle of conservation of energy,
loss in G.P.E. = gain in K.E.

$$mgh - 0 = \frac{1}{2}mv^2 - 0$$

$$v = \sqrt{2gh} = \sqrt{2 \times 9.81 \times 0.70} = 3.7 \text{ m s}^{-1}$$

ii.



T is tension, mg is weight.

- iii. The centripetal force,

$$F_c = \frac{mv^2}{r} = \frac{0.050 \times 3.7^2}{1.50} = 0.456 \text{ N}$$

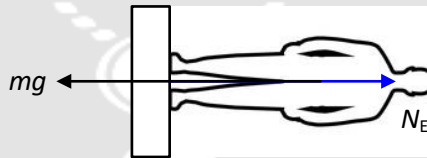
- iv. The ball is moving in a circular path (with centre directly above the ball) hence it must experience an upward acceleration. Since tension T and weight mg are the only forces acting on the ball, $T \neq mg$.

- v. Resultant force provides for centripetal force.

$$T - mg = F_c \Rightarrow T = F_c + mg = 0.456 + 0.050 \times 9.81 = 0.947 \text{ N}$$

9. [YJC/H2/Prelims 2018/P3/2]

a.



- b. The scale reads the normal force N .

At P, since the man is not undergoing circular motion, $N_p = mg$.

At E, since the man is undergoing circular motion, centripetal force is provided by the resultant force

$$mg - N_E = ma_c$$

$$\Rightarrow N_E = mg - ma_c < N_p$$

- c. Rearranging,

$$a_c = \frac{mg - N_E}{m} = \frac{N_p - N_E}{m} = \frac{589.2 - 587.4}{60} = 0.030 \text{ m s}^{-2}$$